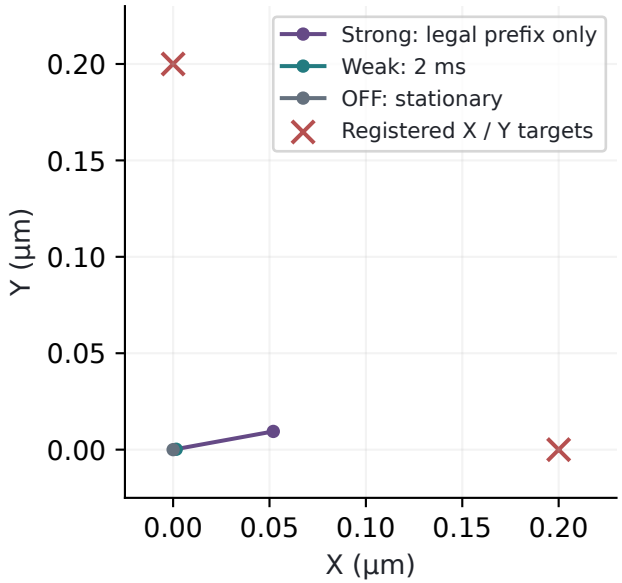
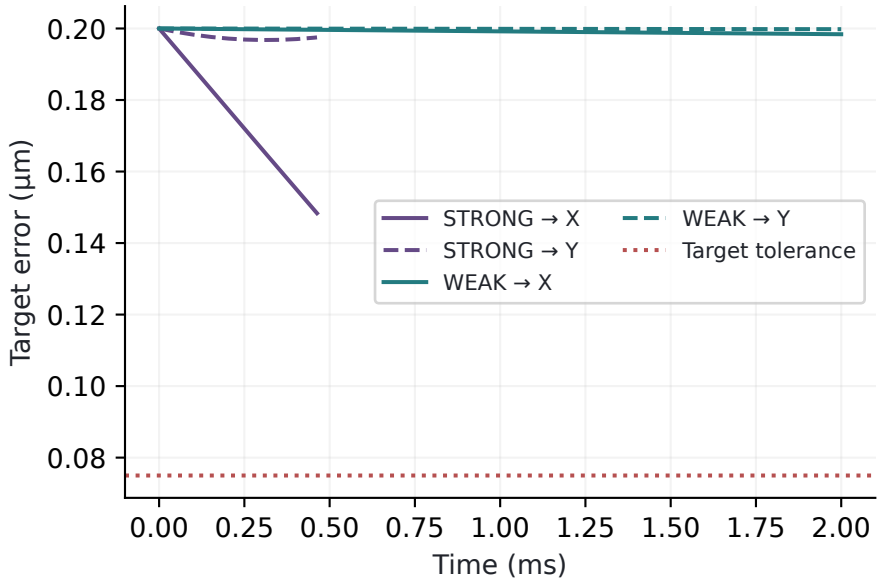
XY navigation: registered targets, actual control and stopping evidence

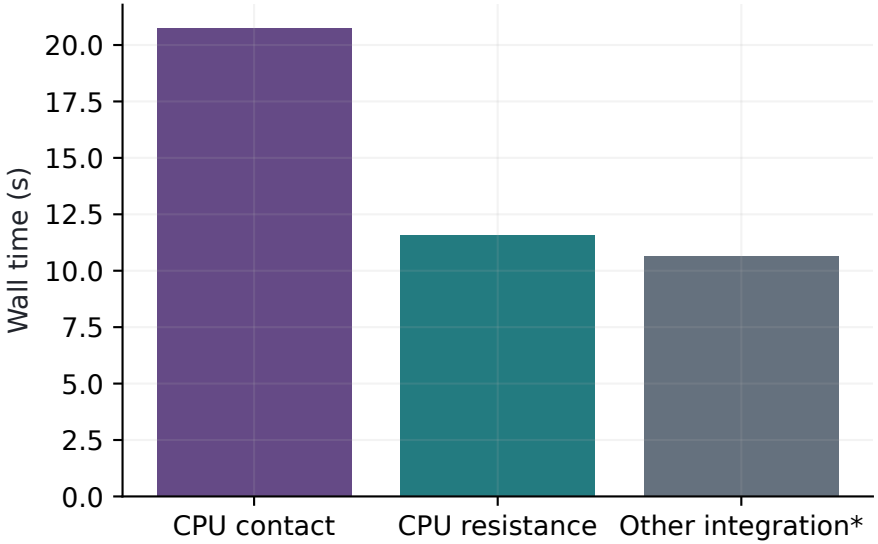
a Actual COM; XY search NOT_RUN



b Error is not target admission



c Measured integration cost



*Includes GPU calls, transfers and evidence; not pure GPU time

d Stop on numerical evidence, not exhausted budget

Scientific numeric processes: 44.86 s
Registered budget: approximately 600 s
Peak process RSS: 537.0 MiB

Stop: true-gap complementarity at 0.464 ms
Strong full 2 ms: FAIL
New XY optimization: NOT_RUN
5 / 10 / 30 ms extensions: NOT_RUN

Next: geometric consistency of contact reactions;
then repeat full-window group admission.
No spring tuning or quality-budget relaxation.